

DIMENSION REDUCTION METHODS

(MHHTK - 28/5/2021)

1 Review on principal component analysis (PCA)

Principal Component Analysis (**PCA**) is a multivariate technique designed to summarize the most important features and relations of k numerical random variables $X_1, \dots, X_k$. **PCA** computes a new set of variables, the principal components $PC_1, \dots, PC_k$, that contain the same information as $X_1, \dots, X_k$ but expressed in a more convenient way. The goal of **PCA** is to retain only a limited number $1 \leq l < k$ of principal components such that they explain most of the information and perform dimension reduction.

1.1 If $X_1, \dots, X_k$ are centred

If $X_1, \dots, X_k$ are centred (That is , $\mathbb{E}[X] = 0, j = 1, \dots, k$), then the principal components **PC** are orthonormal linear combinations of $X_1, \dots, X_k$:

$$PC_j = \sum_{i=1}^p a_{ij} X_i = a_{1j} X_1 + \dots + a_{kj} X_k = \mathbf{a}_j' \mathbf{X}$$

where

- $\mathbf{a}_j = (a_{1j}, \dots, a_{kj})'$
- $\mathbf{X} = (X_1, \dots, X_k)'$
- $\mathbf{a}_i' \mathbf{a}_j = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{if } i \neq j. \end{cases}$

Remark: *PCA computes the principal components in an ordered way: PC_1 is the principal component that explains the most of the information (quantified as the variance) of $X_1, \dots, X_k$, and then the explained information decreases monotonically down to PC_k , the principal component that explains the least information. Precisely:*

$$\text{Var}(PC_1) \geq \text{Var}(PC_2) \geq \dots \geq \text{Var}(PC_k).$$

The covariance matrix $\Sigma := \text{Var}(\mathbf{X})$:

$$\Sigma := \text{Var}(\mathbf{X}) = \mathbf{A}' \mathbf{\Lambda} \mathbf{A}$$

where $\mathbf{\Lambda} = \text{diag}(\lambda_1, \dots, \lambda_k)$ contains the eigenvalues of $\mathbf{\Sigma}$ and $\mathbf{A}$ is the orthogonal matrix that contains the unit-norm eigenvectors of $\mathbf{\Sigma}$ as columns.

The matrix $\mathbf{A}$ gives, thus, the coefficients of the orthonormal linear combinations:

$$\mathbf{A} = \begin{pmatrix} \mathbf{a}_1 & \mathbf{a}_2 & \dots & \mathbf{a}_k \end{pmatrix} = \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1k} \\ a_{21} & a_{22} & \dots & a_{2k} \\ \dots & \dots & \dots & \dots \\ a_{k1} & a_{k2} & \dots & a_{kk} \end{pmatrix}$$

1.2 If $X_1, \dots, X_k$ is not centred

If the data is not centred, the computation of the principal components is done by first subtracting $\mu = \mathbb{E}[\mathbf{X}]$ and then premultiplying with $\mathbf{A}$:

$$\mathbf{PC} = \mathbf{A}'(\mathbf{X} - \mu)$$

with $\mathbf{PC} = (PC_1, PC_2, \dots, PC_k)'$, $\mathbf{X} = (X_1, \dots, X_k)'$, and $\mu = (\mu_1, \dots, \mu_k)'$. Note that, because of (1.3) and (3.7), Note that,

$$\mathbb{V}ar(\mathbf{PC}) = \mathbf{A}'\mathbf{\Lambda}\mathbf{A} = \mathbf{A}'\mathbf{A}\mathbf{\Lambda}\mathbf{A}'\mathbf{A} = \mathbf{\Lambda}.$$

therefore $\mathbb{V}ar(PC_1) = \lambda_j$, $j = 1, 2, \dots, k$ and

$$\mathbb{V}ar(PC_j) \geq \mathbb{V}ar(PC_2) \geq \dots \geq \mathbb{V}ar(PC_k).$$

We can express the random variables $\mathbf{X}$ in terms of $\mathbf{PC}$:

$$\mathbf{X} = \mu + \mathbf{A}\mathbf{PC}$$

which admits an insightful interpretation: the $\mathbf{PC}$ are uncorrelated variables that, once rotated by $\mathbf{A}$ and translated to the location μ , produce exactly $\mathbf{X}$. So, $\mathbf{PC}$ contains the same information as $\mathbf{X}$ but rearranged in a more convenient way because the principal components are centred and uncorrelated between them:

- $\mathbb{E}[PC_i] = 0$
- $\mathbb{C}ov(PC_i, PC_j) = \begin{cases} \lambda_i & \text{if } i = j \\ 0 & \text{if } i \neq j. \end{cases}$

Given the uncorrelation of **PC** , we can measure **the total variance of PC** as

$$\sum_{j=1}^k \text{Var}(PC_j) = \sum_{j=1}^k \lambda_j$$

Consequently, we can define **the proportion of variance explained** by the first l principal components, $1 \leq l < k$, as

$$\frac{\sum_{j=1}^l \lambda_j}{\sum_{j=1}^k \lambda_j}.$$

2 Principal components regression (PCR)

The key idea behind Principal Components Regression (PCR) is to regress the response Y in a set of principal components $PC_1, PC_2, \dots, PC_l$ obtained from the predictors $X_1, \dots, X_k$, where $l < k$. The motivation is that often a small number of principal components is enough to explain most of the variability of the predictors and consequently their relationship with Y . Therefore, we look for fitting the linear model:

$$Y = \alpha_0 + \alpha_1 PC_1 + \alpha_2 PC_2 + \dots + \alpha_l PC_l + \varepsilon \quad (1)$$

The main advantages of PCR are two:

- Multicollinearity is avoided by design: $PC_1, PC_2, \dots, PC_l$ are uncorrelated between them.
- There are less coefficients to estimate (l instead of p), hence the accuracy of the estimation increases.

However, keep in mind that PCR affects the linear model in two fundamental ways:

- *Interpretation of the coefficients is not directly related with the predictors*, but with the principal components. Hence, the interpretability of a given coefficient in the regression model is tied to the interpretability of the associated principal component.
- *Prediction needs an extra step*, since it is required to obtain the scores of the new observations of the predictors in the principal components.

The first point is worth discussing now. The PCR model (1) can be seen as a linear model expressed in terms of the original predictors.

To make this point clearer, let's re-express (1) as

$$Y = \alpha_0 + \mathbf{P}\mathbf{C}'_{1:l}\alpha_{1:l} + \varepsilon \quad (2)$$

where the subindex $1 : l$ denotes the inclusion of the vector entries from 1 to l . Now, we can express the PCR problem(2) in terms of the original predictors:

$$Y = \alpha_0 + \mathbf{A}'_{1:l}(\mathbf{X} - \mu)\alpha_{1:l} + \varepsilon = (\alpha_0 - \mu'\mathbf{A}_{1:l}\alpha_{1:l}) + \mathbf{X}'\mathbf{A}_{1:l}\alpha_{1:l} + \varepsilon = \gamma_0 + \mathbf{X}'\gamma_{1:k} + \varepsilon$$

where

- $\mathbf{A}'_{1:l}$ represents the $\mathbf{A}$ matrix with only its first l columns
- $\gamma_0 := \alpha_0 - \mu'\mathbf{A}_{1:l}\alpha_{1:l}$
- $\gamma_{1:k} := \mathbf{A}_{1:l}\alpha_{1:l}$

In other words, the coefficients $a_1 : a_l$ of the PCR done with l principal components in (1) translate into the coefficients γ_0 and $\gamma_{1:k}$ of the linear model based on the k original predictors

$$Y = \gamma_0 + \gamma_1 X_1 + \dots + \gamma_k X_k + \varepsilon \quad (3)$$

In the sample case, we have that

- $\hat{\gamma}_0 := \hat{\alpha}_0 - \bar{X}'\hat{\mathbf{A}}_{1:l}\hat{\alpha}_{1:l}$
- $\hat{\gamma}_{1:k} := \hat{\mathbf{A}}_{1:l}\hat{\alpha}_{1:l}$

Notice that $\hat{\gamma}$ is not **the least squares estimator** that we use to denote by $\hat{\beta}$, but just the coefficients of the PCR that are associated to the original predictors. Consequently, $\hat{\gamma}$ is useful for the interpretation of the linear model produced by PCR, as it can be interpreted in the same way $\hat{\beta}$ was. Finally, remember that the usefulness of PCR relies on how well we are able to reduce the dimensionality of the predictors and the veracity of the assumption that the l principal components are related with Y .